https://Antoine-dSG.github.io/root$_{s}ystemhttps://github.com/Antoine-dSG/root_{s}ystemhttps://Ant$

Root systems

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Theorem 1. *The 8-tuple* $\left(\mathbb{Z}, \mathbb{Z}^l, \left(\mathbb{Z}^l\right)^*, \mathcal{L}, I, \alpha, \alpha^\vee, s\right)$ *is a root pairing of* A_l*, where:*

1. $\mathcal{L}$ *is the dot product on* $\mathbb{Z}^l \times \left(\mathbb{Z}^l\right)^*$
2. $I = \{$*signed intervals on the set* $\{1, ..., n\}\} = I^+ \cup I^-$
3. $s : I \to Perm(I)$ *where for* $J \in I$*,* $s_J = s_{-J}$*,* $s_J(-K) = -s_J(K)$*, defined as* $s_J(K) =$

$$\begin{cases} J \cup K & \text{if } J \cap K = \emptyset \wedge J \cup K \in I \\ K & \text{if } J \cap K = \emptyset \wedge J \cup K \notin I \\ K \setminus J & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \notin I \\ -K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K = \emptyset \\ -(J \setminus K) & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \notin I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \nsubseteq J \wedge J \not\subset K \end{cases}$$

Equivalently (for use in Lean),

1. $I = \{(i, j, \varepsilon) | 1 \leq i \leq j \leq l \in \mathbb{Z}, \varepsilon = \pm 1\}$
2. $\alpha : I \hookrightarrow \mathbb{Z}^l$, $(i, j, \varepsilon) \mapsto \varepsilon \left(\sum_{k=i}^{j} A_l e_k\right)$ *(where* A_l *is the Cartan Matrix of* A_l*)*
3. $\alpha^\vee : I \hookrightarrow \left(\mathbb{Z}^l\right)^*$, $(i, j, \varepsilon) \mapsto \varepsilon \left(\sum_{k=i}^{j} e_k^\top\right)$
4. $s : I \to Perm(I)$ *where* $s(i, j, 1) = s(i, j, -1)$, $s(i, j, \varepsilon)(k, l, 1) = (p, q, \eta)$ *means that* $s(i, j, \varepsilon)(k, l, -1) = (p, q, -\eta)$*, and*

$$s(i,j,1)(k,l,1) = \begin{cases} (k, l, -1) & \text{if } (i, j) = (k, l) \\ (l+1, j, -1) & \text{if } i = k, j > l \\ (j+1, l, 1) & \text{if } i = k, j < l \\ (i, k-1, -1) & \text{if } i < k, j = l \\ (k, i-1, 1) & \text{if } i > k, j = l \\ (k, j, 1) & \text{if } i = l+1 \\ (i, l, 1) & \text{if } k = j+1 \\ (k, l, 1) & \text{otherwise} \end{cases}$$

Proof. ☐